

Disclaimer

- This training resource was developed to be used with the CZ ID platform
- In October, 2026, stewardship of CZ ID transitioned to UCSF and the platform was renamed to SeqtoID
- These training materials are not being actively maintained and may not reflect differences between CZ ID and SeqtoID, subsequent changes to the platform, or current methods and best practices in metagenomic sequencing and analysis



Quality Control Overview and Interpretation

Outline

- QC metric definitions
- How QC metrics help inform wet lab experiments and data interpretation
- Perform QC within CZ ID (**look at your data!**)

Quality Control

- Were there enough reads?
- Did my samples pass QC filtering? (basecall quality, length, complexity)
- How many reads were removed during data preprocessing? (e.g., host filtering)
- How many reads were used for microbe identification (i.e., passed filters)?

CZ ID QC Metrics

Total Reads: Total reads uploaded to the pipeline.

Passed QC: Reads that passed quality filters (fastp).

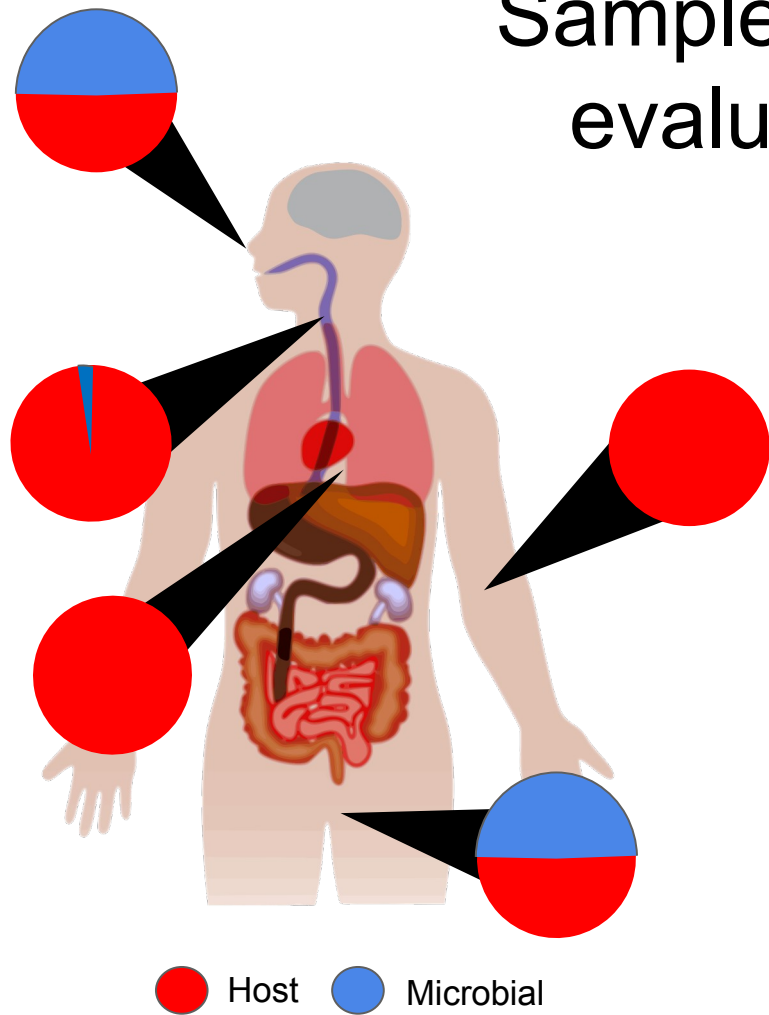
Passed Filters: Reads used for analysis (i.e., reads that passed QC, host/human filtering, and subsampling).

Duplicate Compression Ratio (DCR): Ratio of sequences present prior to duplicate removal vs after duplicate removal (e.g. over-amplification, over-sequencing).

ERCC Reads: Total number of reads mapping to ERCC internal controls (RNA-seq).

Mean insert size: Average length of the nucleotide sequence that is inserted between adapters (human samples only).

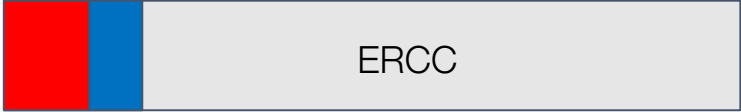
Sample type matters when evaluating QC metrics



Sample Type	Host Signal	Microbial Signal
CSF	> 99%	<< 5% with infection
Tracheal Aspirate	95%	5%
Nasal or oral	50%	50%
Whole blood	> 99%	<< 1% (with infection)
Rectal or Vaginal	50%	50%

*Concepts introduced in detail in previous modules
(Introduction to mNGS; mNGS Study Design)

QC metrics affected by sample type

Sample Type	Typical Distribution 	QC Trend
Stool, NP Swab, Environmental Samples, Sputum, Urine (Higher diversity & DNA/cDNA input)		• Higher microbial reads • Lower DCR
Serum, CSF, Plasma, Negative Controls (Lower diversity & DNA/cDNA input)		• Lower microbial reads • Higher DCR (PCR artifact) • Higher background contamination

QC metrics not affected by sample type

- **% Passed QC (fastp)**

- Should not inherently vary by tissue type
 - Per base quality (Q score)
 - Adapter dimer
 - Length
 - Complexity
- May require knowledge of sample processing for interpretation

- **Total reads**

- Not affected by sample type
- Dependent on input DNA/cDNA and sequencing depth

Quality metrics on the Project page

(Sample Table)

0

Search

Samples 217

0 Selected

☐	Sample	Uploaded On	Total Reads	Passed Filters	Passed QC	DCR	
☐	OPS015_UGD-HIV-PNEU_BAL_RNA_2460_H8... COMPLETE Chaz Langelier Uganda-CEND-pneumonia-cases	2019-07-31 3 months ago	74,775,512	67,602 0.09%	92.10%	1.60	
☐	OPS015_UGD-HIV-PNEU_BAL_RNA_2455_F8... COMPLETE Chaz Langelier Uganda-CEND-pneumonia-cases	2019-07-31 3 months ago	84,091,062	42,754 0.05%	88.16%	1.49	
☐	OPS015_UGD-HIV-PNEU_BAL_RNA_2444_D8... COMPLETE Chaz Langelier Uganda-CEND-pneumonia-cases	2019-07-31 3 months ago	80,180,516	4,533,252 5.65%	95.37%	1.73	
☐	OPS015_UGD-HIV-PNEU_BAL_RNA_2442_B8... COMPLETE Chaz Langelier Uganda-CEND-pneumonia-cases	2019-07-31 3 months ago	82,283,136	16,322 0.02%	81.98%	1.50	
☐	OPS015_UGD-HIV-PNEU_BAL_RNA_2439_P6... COMPLETE Chaz Langelier Uganda-CEND-pneumonia-cases	2019-07-31 3 months ago	94,471,904	2,518 <0.01%	86.47%	2.04	
☐	OPS015_UGD-HIV-PNEU_BAL_RNA_2432_N6... COMPLETE Chaz Langelier Uganda-CEND-pneumonia-cases	2019-07-31 3 months ago	92,331,992	175,034 0.19%	86.72%	1.62	
☐	OPS015_UGD-HIV-PNEU_BAL_RNA_2423_L6... COMPLETE Chaz Langelier Uganda-CEND-pneumonia-cases	2019-07-31 3 months ago	95,485,476	6,288 0.01%	86.81%	2.05	

Project-level QC

Uganda-CEND-pneumonia-cases

Public project ⓘ 5 members

Share

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Search My Data...

Samples 217



- TAXON ▾
- LOCATION ▾
- TIMEFRAME ▾
- VISIBILITY ▾
- HOST ▾
- SAMPLE TYPE ▾

Metagenomics 217 Consensus Genomes 0



0 Selected



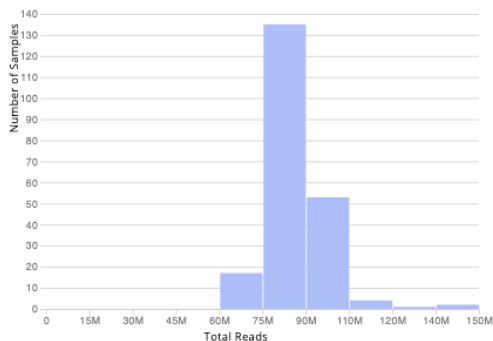
5 samples are still being processed.



Showing 212 of 217 samples ⓘ

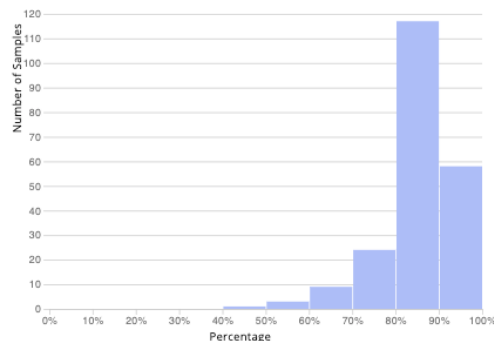
Do my samples have enough total reads?

Total Reads ⓘ



Do my samples have enough quality reads?

Passed QC ⓘ



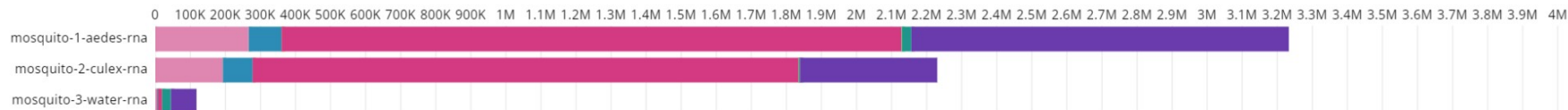
Project level QC - reads lost

Mosquito

Reads Lost ⓘ

Validate Input Filter low quality Filter length Filter low complexity Ercc Filter host (Bowtie2) Filter host (Hisat2) Filter human (Bowtie2) Filter human (Hisat2) Subsampling Passed Filters

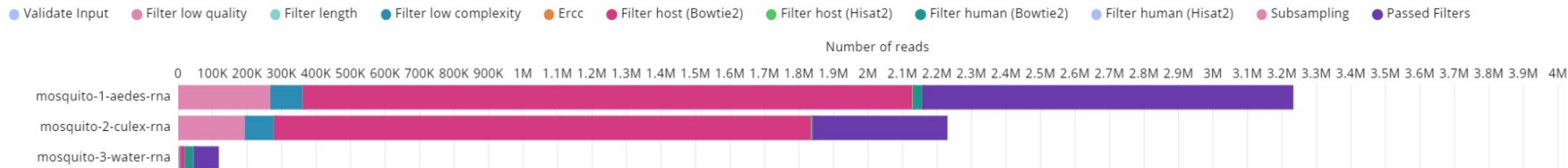
Number of reads



Project level QC - reads lost

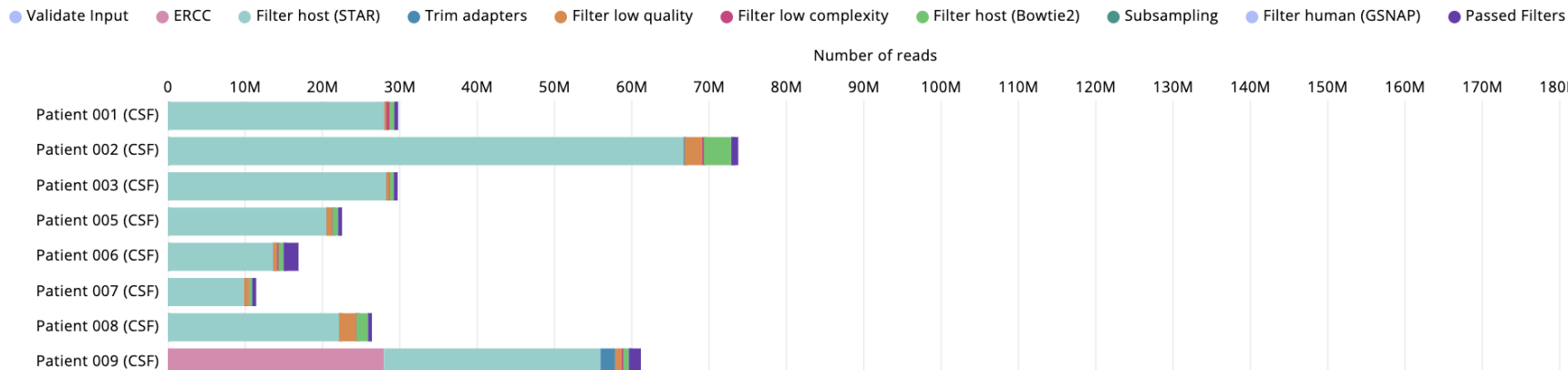
Mosquito

Reads Lost ⓘ



CSF

Reads Lost ⓘ



Reads Lost Interpretation

Pipeline Step	Function	If Many Reads Lost During this Step...
fastp	Filter length	Many short reads (possible adapter dimers, degradation, and/or over-fragmentation).
	Filter low quality reads	Many reads of low-quality (low phred scores, GC content, over-clustering)
	Filter low complexity reads	Reads are relatively low-complexity (contain homopolymer nucleotide runs or simple sequence repeats)
Bowtie2, HISAT2	Filter host reads	Large proportion of reads mapped to the host genome
Bowtie2, HISAT2	Filter human reads (regardless of host)	Large proportion of reads mapped to the human genome

Sample-level QC

PIPELINE v3.7, NT/NR: 2018-12-01 | processed 3 months ago

Uganda-CEND-pneumonia-cases >

OPS015_UGD-HIV-PNEU_BAL_RNA_2460_H8_S16 ▾

[Sample Details](#)

Report Antimicrobial Resistance

Taxon name **Name Type:** Scientific ▾ **Background:** RR038 NextSeq H2O ▾ **Categories** ▾ **Threshold F** ▾

Min Contig Size: 4 ▾

1767 rows passing the above filters, out of 1890 total rows.

> Taxon	Score ^	Z ^	rPM ^	r ^	contig ^
> <i>Faecalibacterium</i> (1 bacterial species)	573,420	100.0 100.0	0.1 57.2	9 4,278	3
> <i>Veillonella</i> (25 bacterial species)	514,804	15.5 40.8	113.9 145.4	8,514 10,867	21
> <i>Megasphaera</i> (6 bacterial species)	421,451	99.0 100.0	53.8 4.5	4,019 340	4
> <i>Chromobacterium</i> (2 bacterial species)	295,870	-0.6 100.0	0.0 29.6	1 2,212	3
> <i>Campylobacter</i> (5 bacterial species) ● 1	222,970	100.0 100.0	21.6 7.9	1,614 594	4
> <i>Bacteroides</i> (5 bacterial species)	125,450	100.0	19.7	1,476	3

OPS015_UGD-HIV-PNEU_BAL_RNA_2460_H8_S16 ✕

Metadata **Pipeline** Notes

PIPELINE INFO

Pipeline Version v3.7 [View Pipeline Visualization >](#)

Total Reads 74,775,512

ERCC Reads 14,336 (0.02%)

Passed Filters 67,602 (0.09%)

Unmapped Reads 82

Passed Quality Control 92.10%

Compression Ratio 1.60

Date Processed 2019-08-01

READS REMAINING

ERCC SPIKE-IN COUNTS

DOWNLOADS

Sample-level QC

PIPELINE v3.7, NT/NR: 2018-12-01 | processed 3 months ago

Uganda-CEND-pneumonia-cases >

OPS015_UGD-HIV-PNEU_BAL_RNA_2460_H8_S16 ▾

[Sample Details](#)

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OPS015_UGD-HIV-PNEU_BAL_RNA_2460_H8_S16 ✕

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ERCC SPIKE-IN COUNTS

DOWNLOADS

Filtering Step	Reads Remaining	% Reads Remaining
Validate Input	3,236,050	100.00%
Fastp Qc	2,872,538	88.77%
Bowtie2 Filter	1,104,606	34.13%
Hisat2 Filter	1,102,726	34.08%
Bowtie2 Human Filter	1,076,524	33.27%
Hisat2 Human Filter	1,076,516	33.27%
CZID-dedup	1,076,516 (973,314 unique)	33.27%
Subsample	1,076,516	33.27%

Sample-level QC

PIPELINE v3.7, NT/NR: 2018-12-01 | processed 3 months ago

Uganda-CEND-pneumonia-cases >

OPS015_UGD-HIV-PNEU_BAL_RNA_2460_H8_S16 ▾

[Sample Details](#)

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> Chromobacterium (2 bacterial species)	295,870	-0.6 100.0	0.0 29.6	1 2,212	0
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OPS015_UGD-HIV-PNEU_BAL_RNA_2460_H8_S16 ✕

Metadata Pipeline Notes

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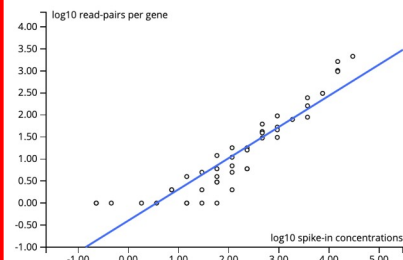
READS REMAINING ▾

ERCC SPIKE-IN COUNTS ▾

DOWNLOADS ▾

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ERCC SPIKE-IN COUNTS



What are ERCC's?

ERCCs = External RNA Controls Consortium

- Pre-mixed, commercially available blend of 92 synthetic RNA transcripts
- Positive control for library preparation, spiked into all samples and water controls
- Can be used to estimate input RNA concentration

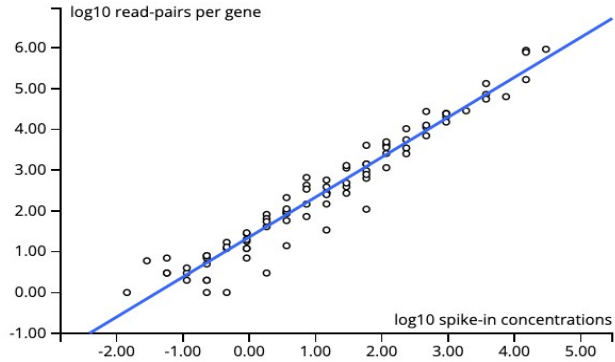
What are ERCC's?

ERCC = External RNA Controls Consortium

- Pre-mixed, commercially available blend of 92 synthetic RNA transcripts
 - Designed to be distinct from RNA found in samples
- Positive control for library preparation, spiked into all samples and water controls
- Can be used to estimate input RNA concentration
 - If we know # of pg of ERCC input and we get out # of reads of ERCC, then we can calculate a (reads per pg of input) factor

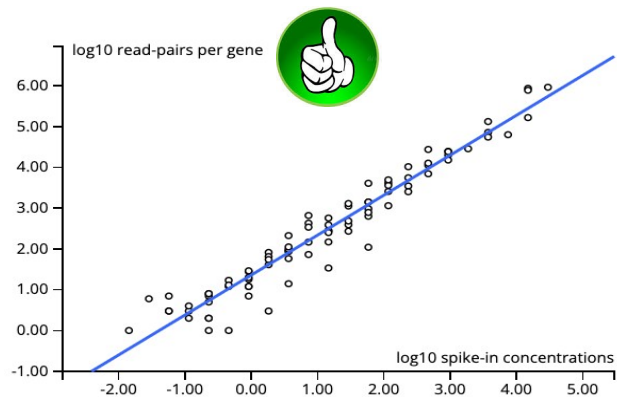
ERCC Plot Examples

ERCC SPIKE-IN COUNTS

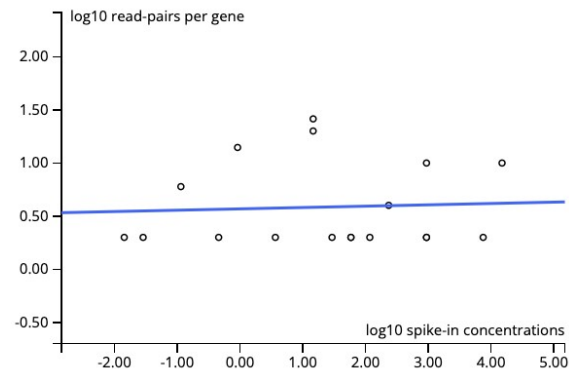


ERCC Plot Examples

ERCC SPIKE-IN COUNTS

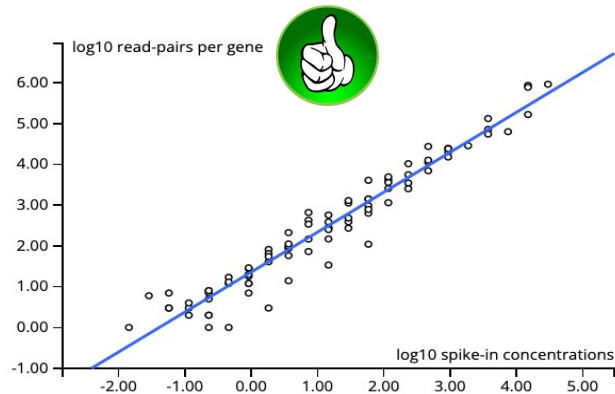


ERCC SPIKE-IN COUNTS

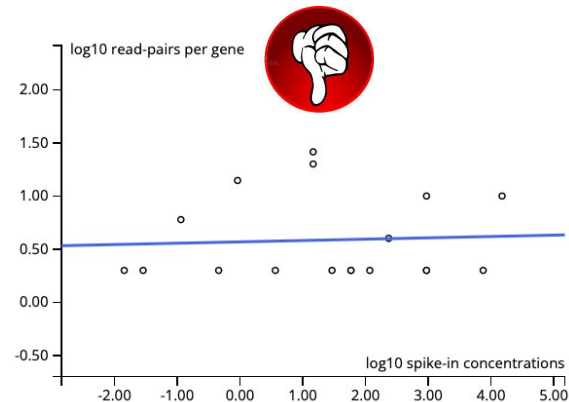


ERCC Plot Examples

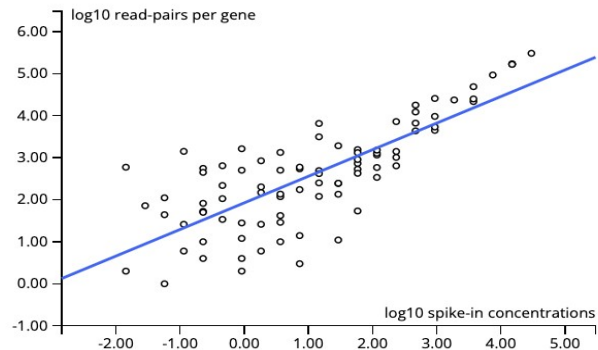
ERCC SPIKE-IN COUNTS



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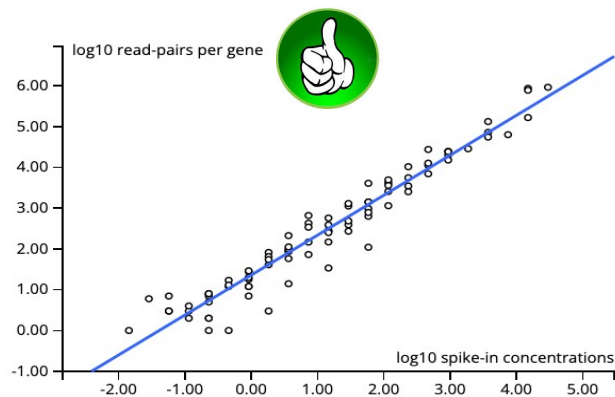


ERCC SPIKE-IN COUNTS

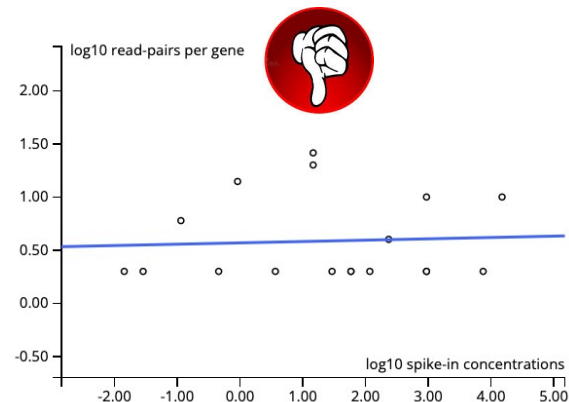


ERCC Plot Examples

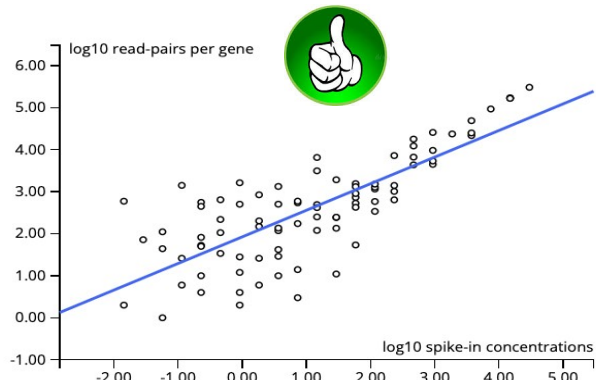
ERCC SPIKE-IN COUNTS



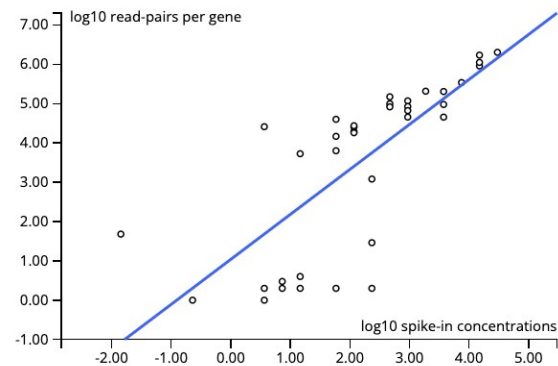
ERCC SPIKE-IN COUNTS



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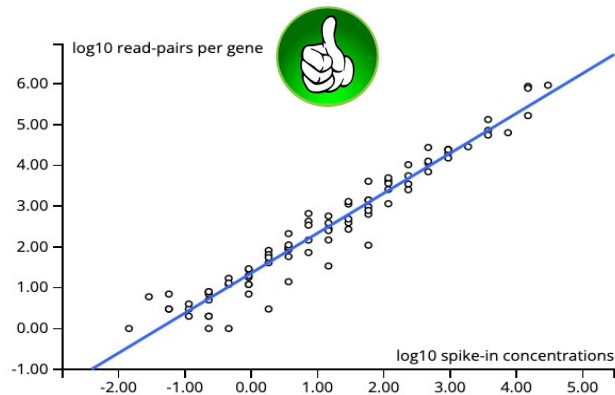


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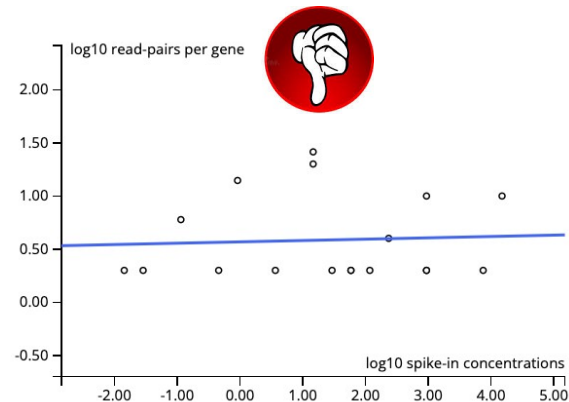


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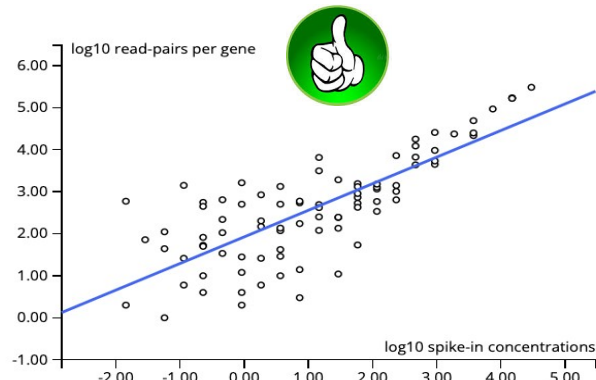
ERCC SPIKE-IN COUNTS



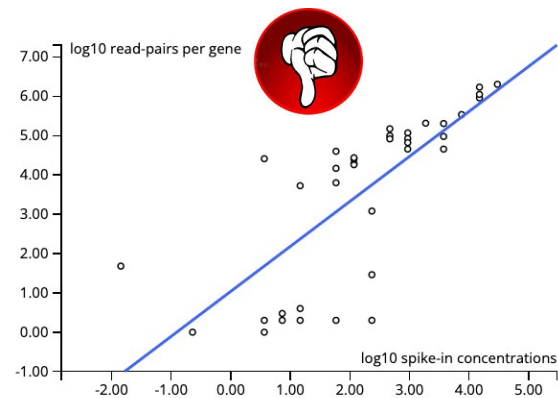
ERCC SPIKE-IN COUNTS



ERCC SPIKE-IN COUNTS



ERCC SPIKE-IN COUNTS



Let's perform QC together!

Current Activity: CZ ID Training Reference Tables & Worksheets_Illumina (pg 5-10)

Next Slides: DL6-I_Sample Report Metrics for Illumina